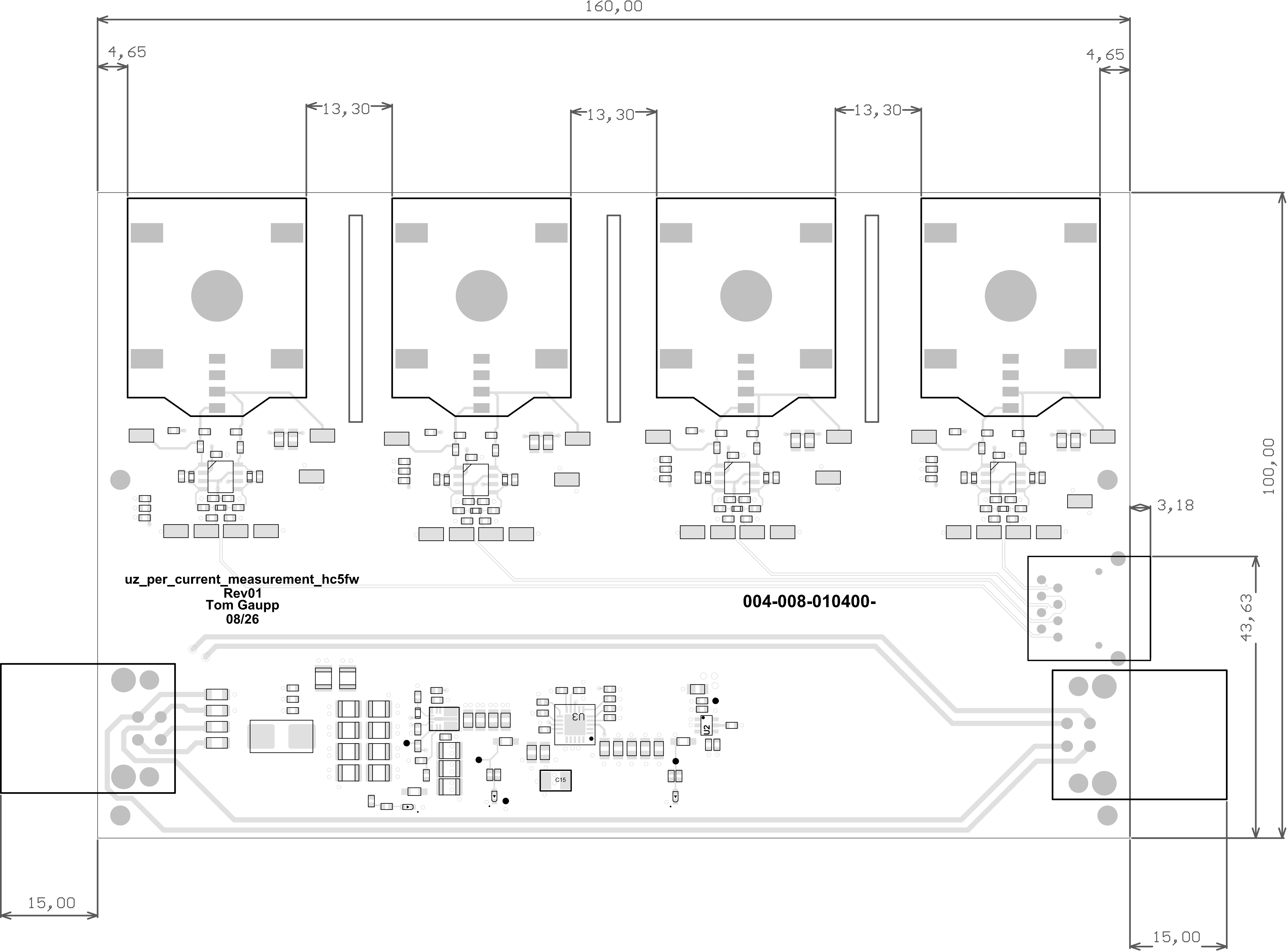




160,00

4,65

4,65

13,30

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13,30

uz_per_current_measurement_hc5fw
Rev01
Tom Gaupp
08/26

004-008-010400-

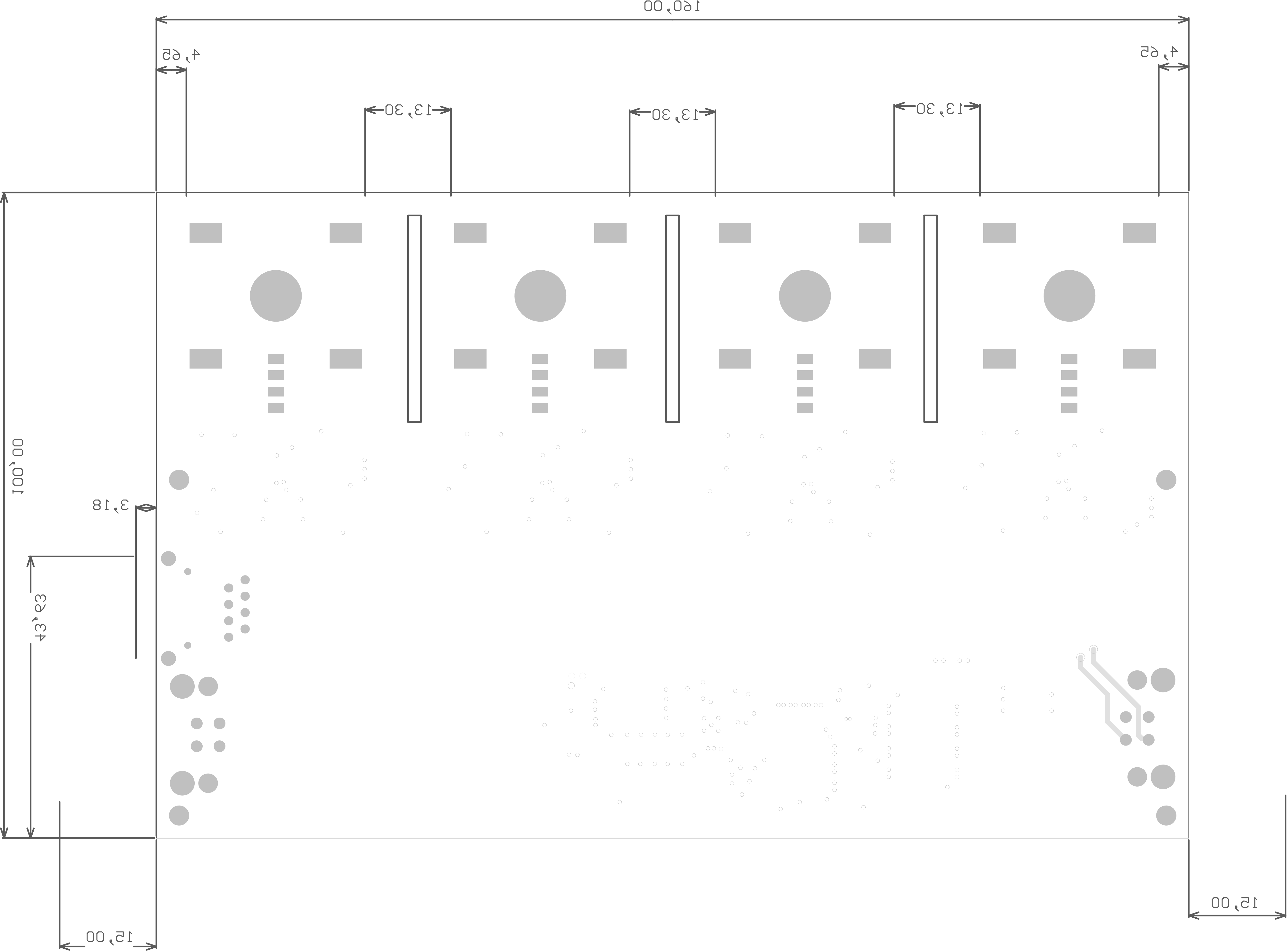
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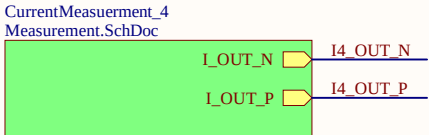
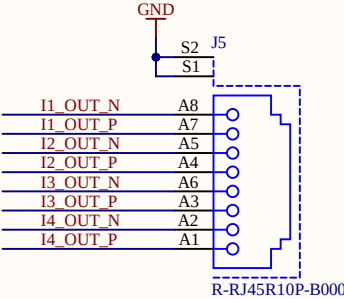
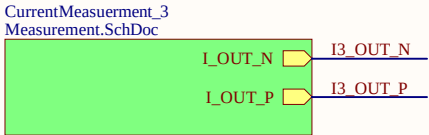
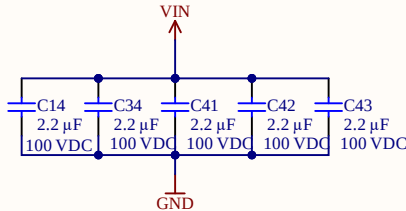
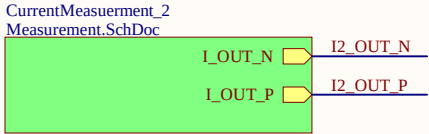
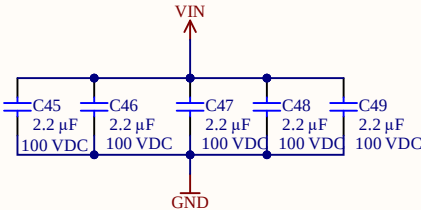
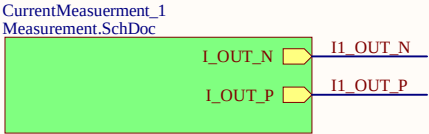
100,00

43,63

15,00

15,00





Power_Supply
Versorgungsspannung



U_Spannungswandlung
Spannungswandlung.SchDoc



LOGO1



UZ Logo

INFO1

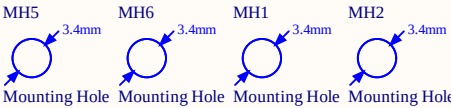
Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

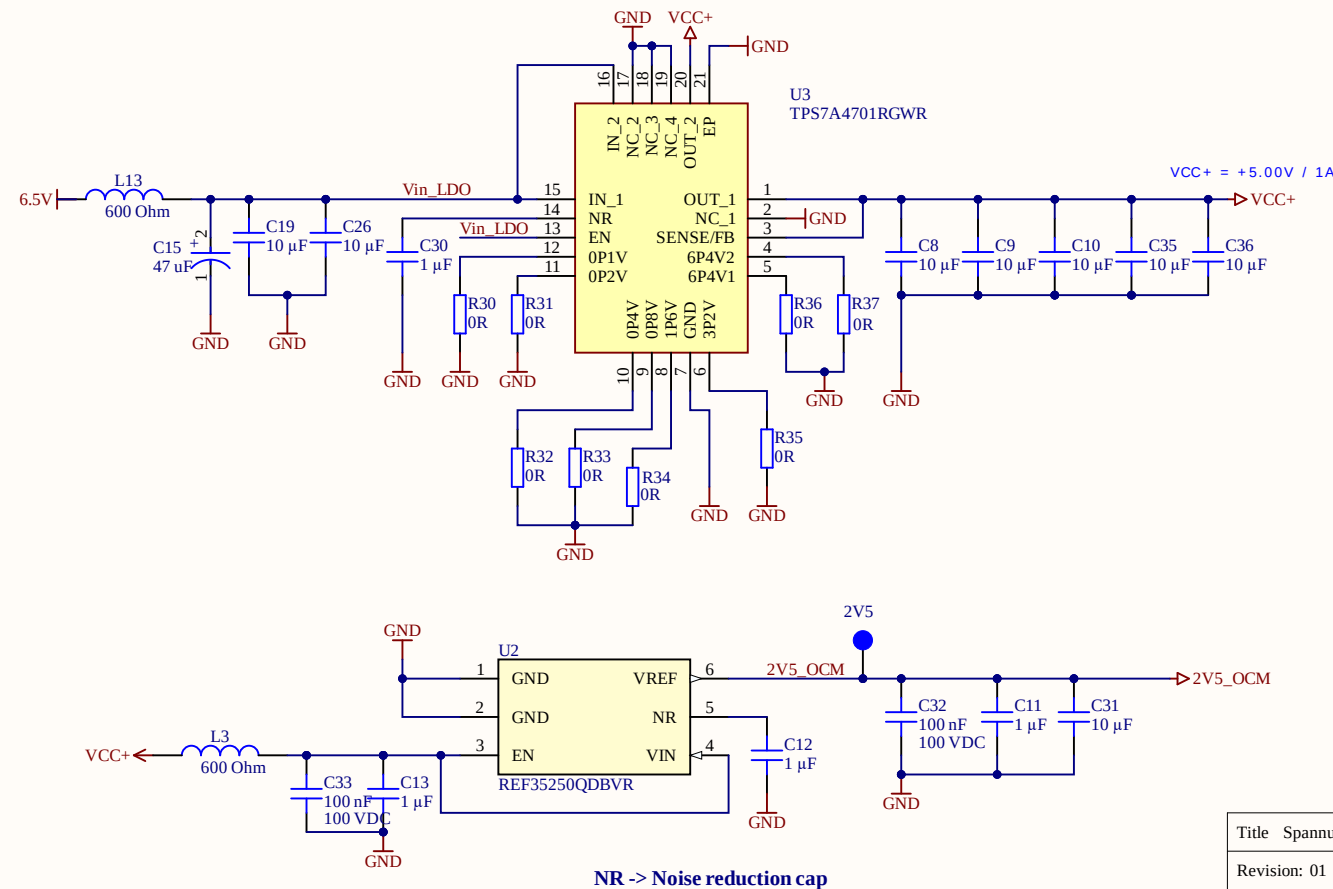
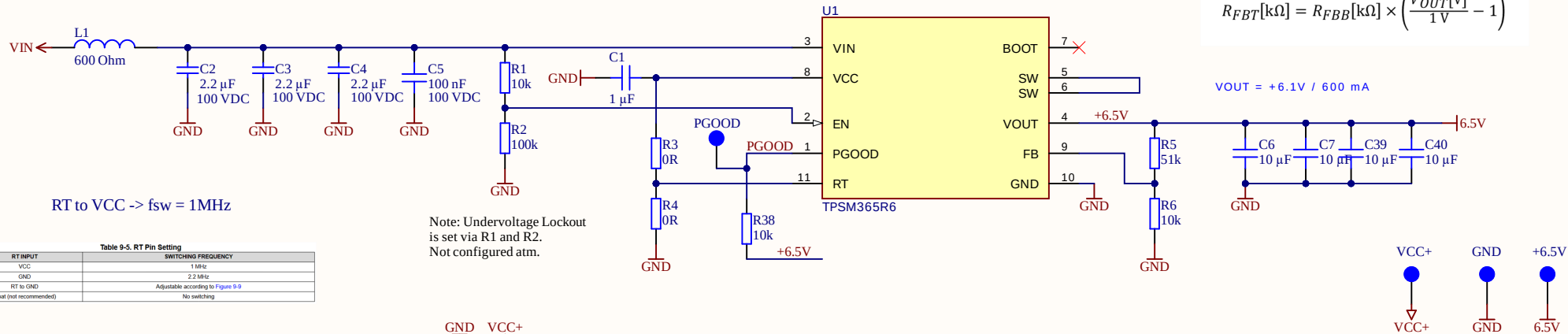
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ZC_PPP
ProjectRevision
ZC_vv

ZC Serialnumber



Title TopSheet.SchDoc		UltraZohm www.ultrazohm.com Date: 19/08/2026 Sheet 1 of 7 
Revision: 01	Design Engineer: Tom Gaupp	
Project: uz_per_current_measurement_hc5fw.PrjPcb		

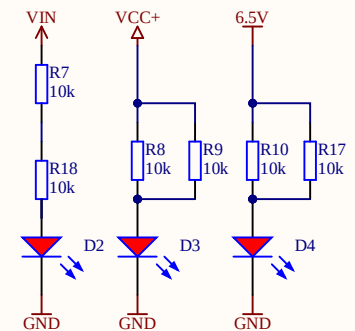
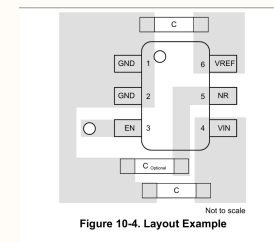


7.5.1 ANY-OUT Programmable Output Voltage

Both devices can be used in ANY-OUT mode. For ANY-OUT operation, the TPS7A4700 and TPS7A4701 do not use external resistors to set the output voltage, but use device pins 4, 5, 6, 8, 9, 10, 11, and 12 to program the regulated output voltage. Each pin is either connected to ground (active) or is left open (floating). The ANY-OUT programming is set by Equation 2 as the sum of the internal reference voltage ($V_{REF} = 1.4V$) plus the accumulated sum of the respective voltages assigned to each active pin; that is, 100 mV (pin 12), 200 mV (pin 11), 400 mV (pin 10), 800 mV (pin 9), 1.6 V (pin 8), 3.2 V (pin 6), 6.4 V (pin 5), or 6.4 V (pin 4). Table 1 summarizes these voltage values associated with each active pin setting for reference. By leaving all program pins open, or floating, the output is thereby programmed to the minimum possible output voltage equal to V_{REF} .

ANY-OUT PROGRAM PINS (Active Low)	ADDITIVE OUTPUT VOLTAGE LEVEL
Pin 4 (OPV2)	6.4 V
Pin 5 (OPV1)	6.4 V
Pin 6 (OP2)	3.2 V
Pin 8 (1P6)	1.6 V
Pin 9 (OP6)	800 mV
Pin 10 (OP4)	400 mV
Pin 11 (OP2)	200 mV
Pin 12 (OP1)	100 mV

$$V_{out} = V_{ref} + \text{SUM(ANY-OUT Pin)} = 1.4V + (3.2V (\text{PIN6}) + 0.4V (\text{PIN10})) = 5.0V$$



Title Spannungswandlung.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_hc5fw.PrjPcb

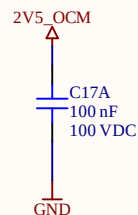
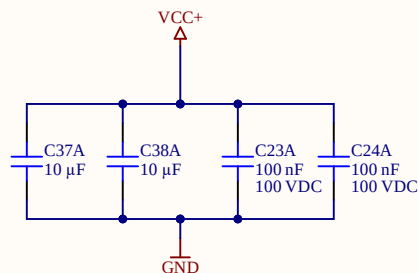
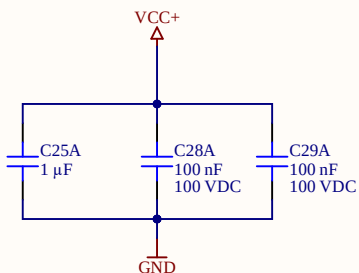
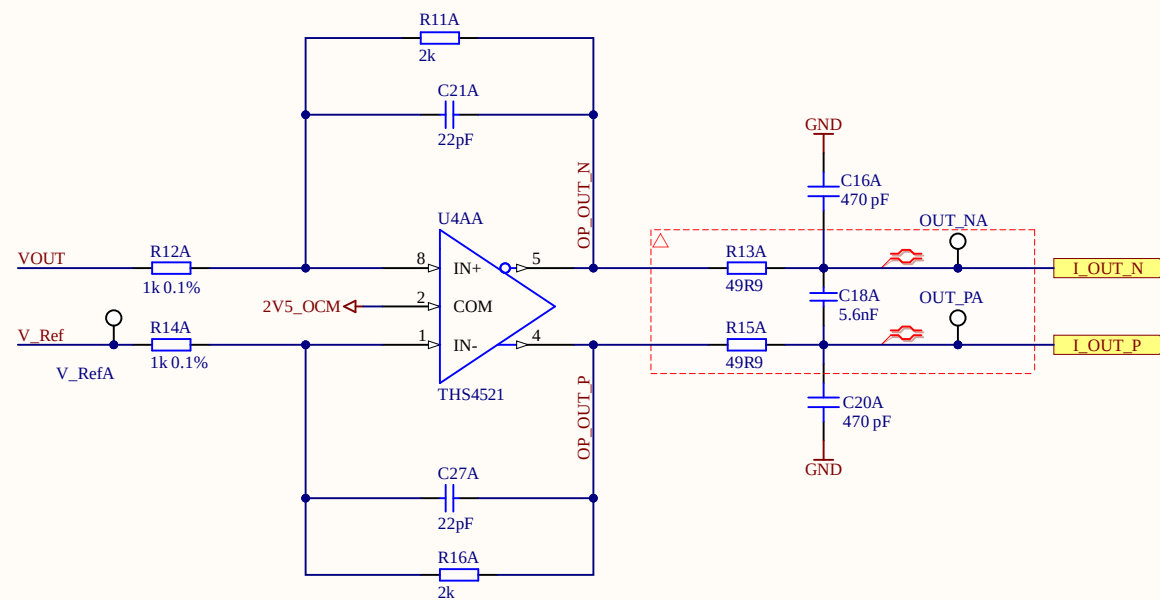
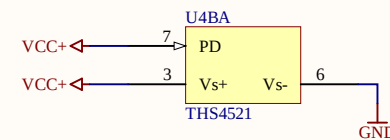
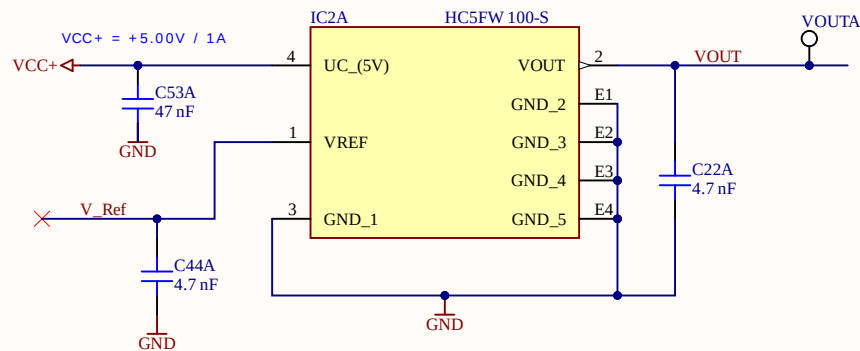
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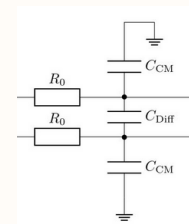
Sheet 2 of 7





$f_{\text{cut,off}} = 273 \text{ kHz}$

$$f_{\text{aliasing, -3dB}} = \frac{1}{2\pi R_0(2C_{\text{Diff}} + C_{\text{CM}})}$$



Title Measurement.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_hc5fw.PrjPcb

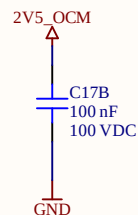
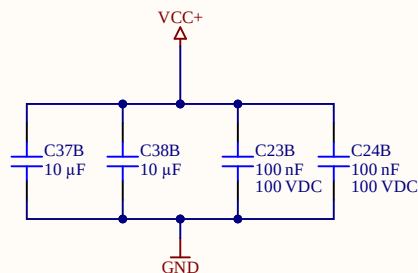
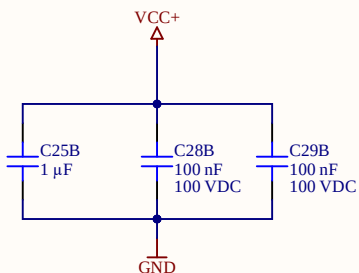
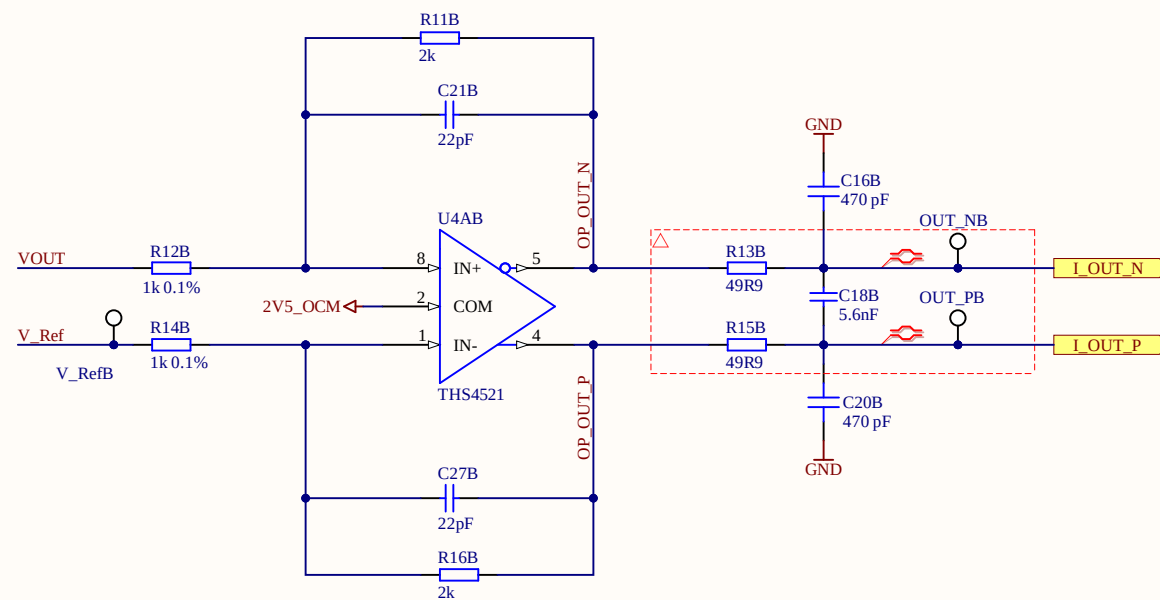
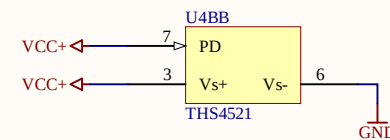
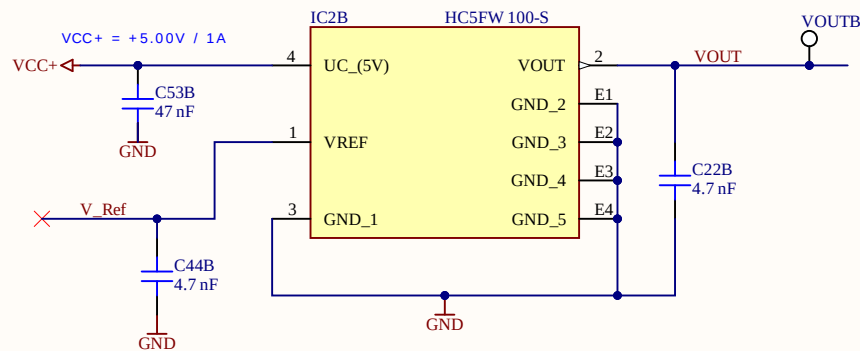
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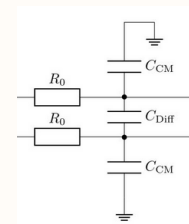
Sheet 3.1 of 7





$f_{\text{cut,off}} = 273 \text{ kHz}$

$$f_{\text{aliasing, -3dB}} = \frac{1}{2\pi R_0(2C_{\text{Diff}} + C_{\text{CM}})}$$



Title Measurement.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_hc5fw.PrjPcb

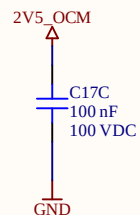
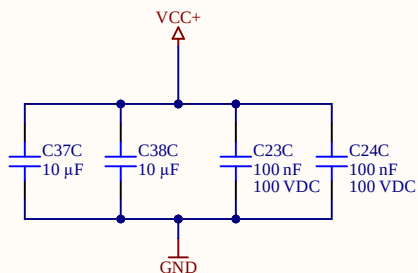
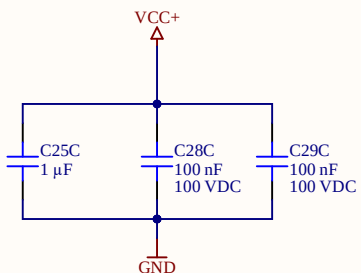
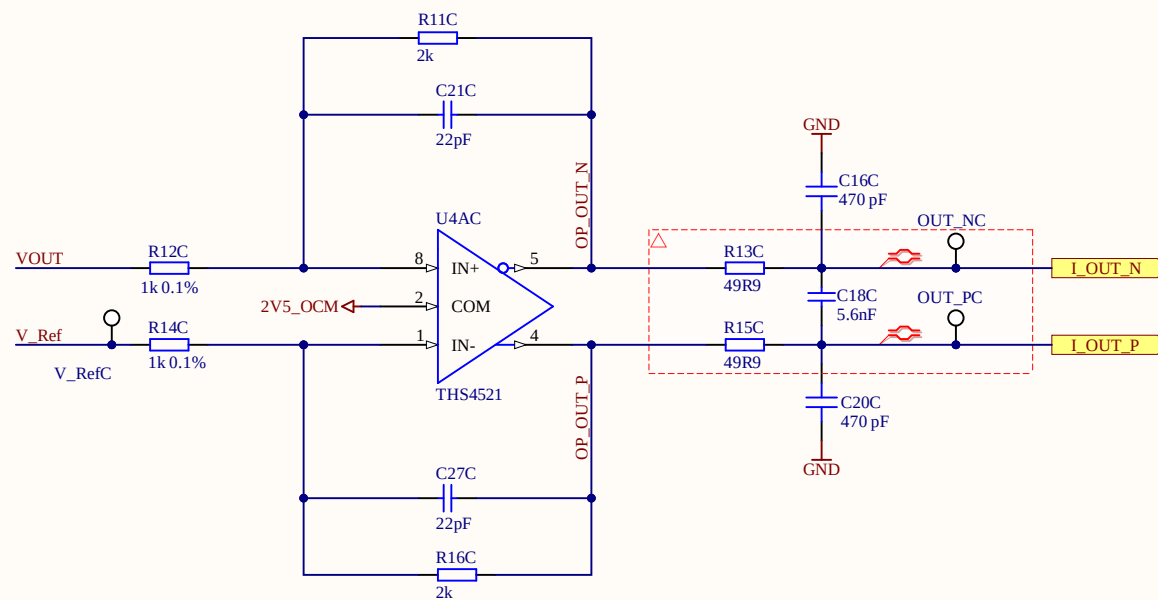
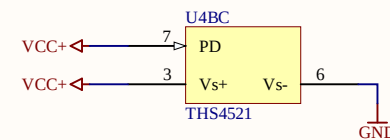
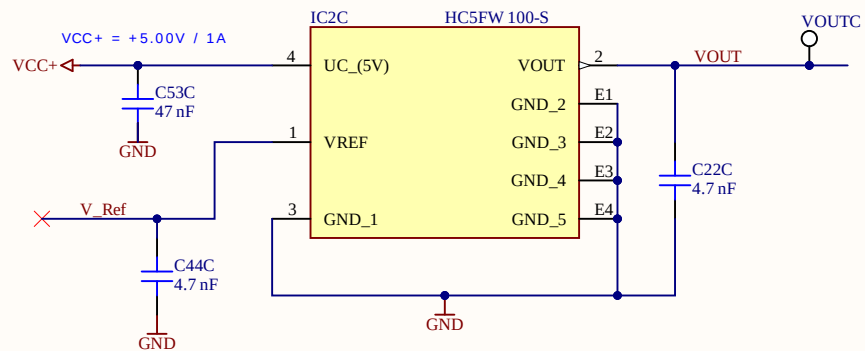
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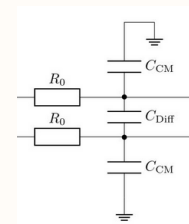
Sheet 3.2 of 7





$f_{\text{cut,off}} = 273 \text{ kHz}$

$$f_{\text{aliasing, -3dB}} = \frac{1}{2\pi R_0 (2C_{\text{Diff}} + C_{\text{CM}})}$$



Title Measurement.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_hc5fw.PrjPcb

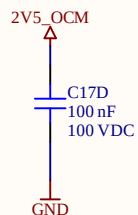
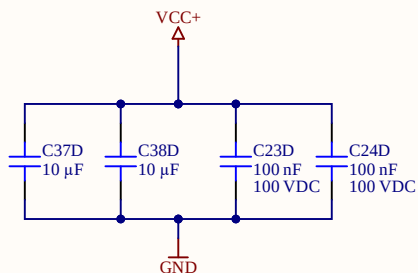
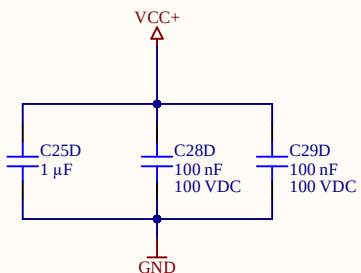
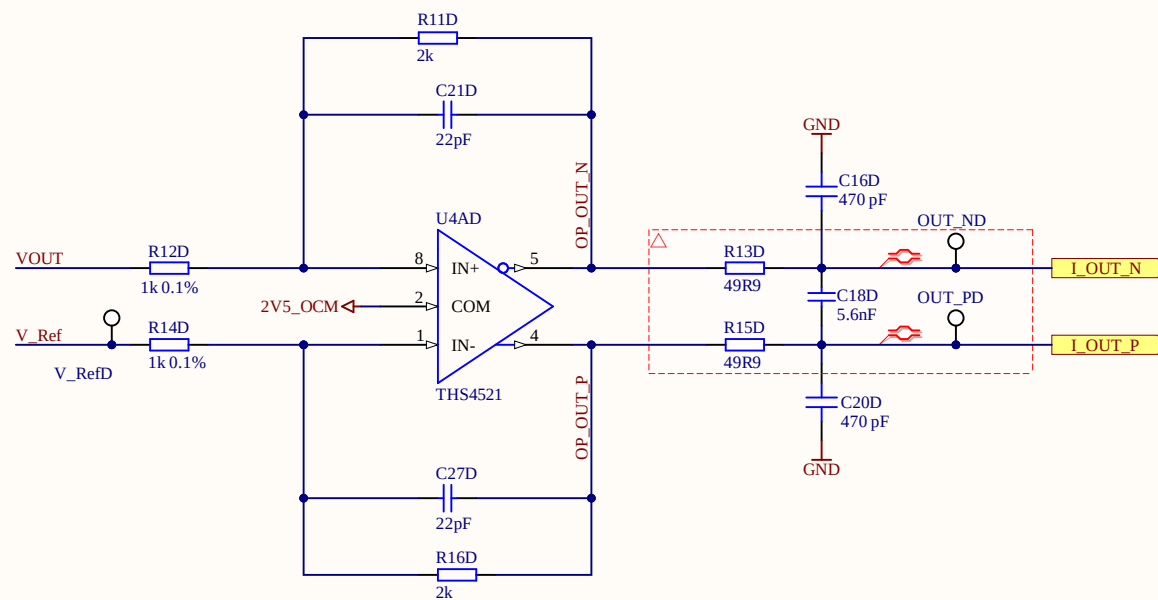
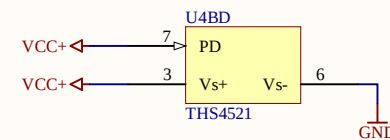
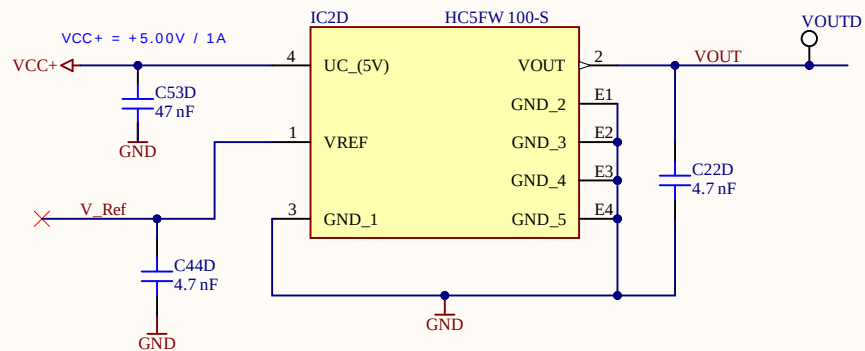
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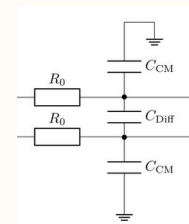
Sheet 3.3 of 7





$$f_{\text{cut,off}} = 273 \text{ kHz}$$

$$f_{\text{aliasing, -3dB}} = \frac{1}{2\pi R_0(2C_{\text{Diff}} + C_{\text{CM}})}$$



Title Measurement.SchDoc

Revision: 01

Design Engineer: Tom Gaupp

Project: uz_per_current_measurement_hc5fw.PrjPcb

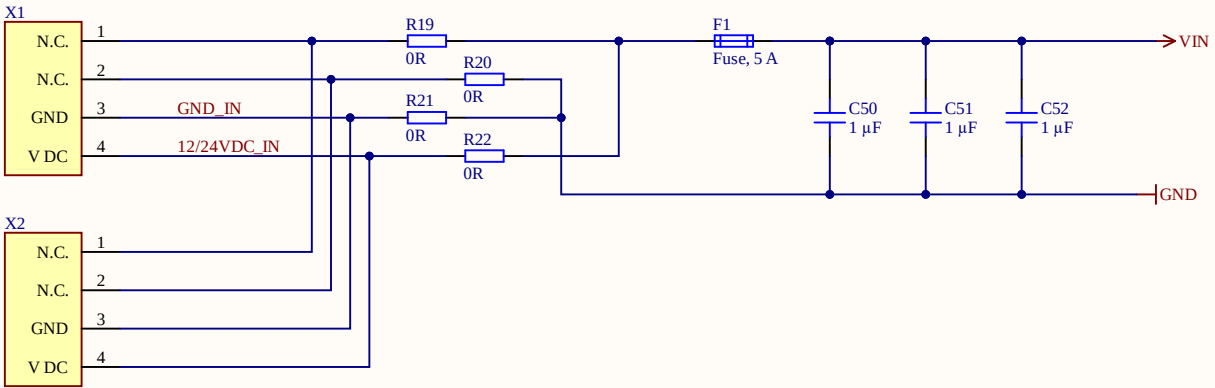
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Date: 19/08/2026

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△ @X1 & @X2 (M12 Buchsen für Stromversorgung):
Pin 1 auf Pin 4 gebrückt und
Pin 2 auf Pin 3 gebrückt

△ Alle Pins über 0R Widerstände
verbunden